

STOCK PRICE TREND, VOLATILITY & RETURN ANALYSIS

SHOPIFY STOCK DATASET — WEEK 4 REPORT

OBJECTIVE

The Shopify stock dataset (2015–2025) was analyzed using Python, Matplotlib and Seaborn to examine long-term price trends, trading volume, moving-average behavior, and the volatility of daily returns. Line plots, a scatter plot, a histogram, and a KDE plot were used to uncover patterns useful for understanding stock risk and performance.

ANALYSIS AND KEY FINDINGS

Analysis	Key Finding	Business Insight
Price Trend (Open/High/Low/Close)	Close price rose from ~\$2.57 (May 2015) to ~\$94.85 (Mar 2025), a ~37x increase over the decade.	Long-term uptrend supports a buy-and-hold growth narrative.
Trading Volume Over Time	Volume spikes coincide with sharp price moves, especially in early trading history.	Volume can act as an early signal of unusual price activity.
Moving Averages (SMA-20 vs SMA-50)	Close price oscillates around both SMAs; crossovers mark shifts in short-term trend.	SMA crossovers are useful, simple trend-following signals.
Volume vs Daily Price Change	No strong linear relationship — high volume days occur across both small and large price changes.	Volume alone is a weak predictor of the size of price moves.
Daily Return Distribution	Mean daily return $\approx 0.09\%$, variance ≈ 9.73 , standard deviation $\approx 3.12\%$.	Returns are centered near zero but carry meaningful day-to-day risk.
High-Volatility Days (90th percentile)	247 of 2,469 trading days (~10%) had absolute returns beyond the 90th-percentile threshold, with swings up to $\pm 10\%+$.	A recurring subset of high-volatility days drives most tail risk.

MAJOR BUSINESS INSIGHTS

- The stock shows strong long-term appreciation, but with substantial short-term volatility along the way.
- Moving averages (SMA-20, SMA-50) smooth out noise and highlight medium-term trend direction.
- Trading volume spikes do not reliably predict the direction or size of price change.
- About 1 in 10 trading days accounts for the bulk of extreme price swings.
- Risk (measured via return variance/std. dev.) should be assessed alongside average return, not in isolation.

CONCLUSION

Visual and statistical analysis of the Shopify dataset shows a stock with strong long-term growth but notable short-term volatility. Combining trend indicators (moving averages) with risk measures (return variance and high-volatility day detection) gives a more complete picture than price charts alone, supporting more informed, risk-aware trading and monitoring decisions.